

Corner Case Detection and Generation for Autonomous Driving: An Overview

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Abstract—Safety concerns remain one of the most significant obstacles to the large-scale deployment and continued advancement of autonomous driving (AD) systems. A major underlying cause of many safety-related incidents in AD systems is suboptimal or erroneous decision-making when the vehicle encounters corner cases (CCs)—rare, unexpected, or extreme situations that fall outside typical operating conditions. Although recent advances in artificial intelligence have driven substantial progress in both autonomous driving and corner-case research, the field still lacks a coherent conceptual foundation and a systematic, widely accepted categorization of CCs. In this survey, we address this gap by offering a structured review of the existing corner-case literature along three key dimensions: understanding, detection, and generation. The key contribution is a three-level classification of corner cases—spanning data-level, model-level, and semantic-level CCs—that clarifies and disambiguates competing definitions and perspectives on AD corner cases. Building on this framework, we examine simulator-based methods for corner-case selection and data generation, and we identify open challenges, promising directions, and potential solutions to more effectively handle corner cases in autonomous driving systems.

Index Terms—Corner case, autonomous driving, out-of-distribution, generated data set.

I. INTRODUCTION

WITH advances in artificial intelligence, significant progress has been made in autonomous vehicles, enhancing perception capabilities and refining decision-making algorithms. However, safety concerns surrounding autonomous driving have garnered growing attention amid the widespread deployment of AI-powered algorithms

in real-world traffic environments [1], [2], [3], [4], [5], [6]—particularly in safety-critical scenarios such as adverse weather conditions [7], [8], adversarial attacks [9], [10], [11], and unseen driving situations [12], [13], [14], [15], [16], [17]. Among the core safety challenges for autonomous driving, rare and safety-critical incidents in the perception stage are of paramount importance: failures in environmental sensing and scene understanding can directly propagate to the downstream planning and control module. This survey, therefore, focuses on corner cases in the perception subsystem of autonomous driving architectures, specifically investigating how in-vehicle sensors acquire, preprocess, and analyze environmental information under rare, unexpected, and hard-to-handle operating conditions.

Existing literature on autonomous driving surveys have already underscored the significance of simulation, diverse data distributions, and safety-critical scenario validation, including autonomous driving policy learning [18], [19], open-source simulators [20], [21], virtual testing tools (encompassing simulators, datasets, and competitive benchmarks) [22], [23], data-driven traffic simulation [24], and foundation-model-based scenario generation and analysis [25]. These works provide valuable contextual insights for autonomous driving validation and long-tail scenario analysis. Nevertheless, they primarily focus on policy learning, simulation platforms, testing pipelines, traffic evolution, or scenario generation/analysis in a broad sense, and do not offer a dedicated survey that makes perception-stage corner cases its primary research focus. This research gap highlights the need for a clearer and more focused synthesis of the literature on perception-oriented corner cases in autonomous driving.

A. Importance of Corner Case for Driving Safety

In autonomous driving research, corner cases (CC) refer to rare and unexpected situations that fall outside normal operating conditions and the usual parameters of perception systems. They are crucial for validating the safety of autonomous vehicle perception modules, which can fail in scenarios not represented in their training data, potentially leading to accidents due to object misclassification, missed obstacles, or inaccurate distance estimation. Typical examples include pedestrians abruptly stepping into the roadway, headlight glare impairing camera visibility, and unusual road hazards such as fallen tree branches. Analyzing these cases enables researchers

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to design methods that reduce their negative impact on sensor interpretation and environment modeling.

Nowadays, there is no consensus on the definition of the term *corner case*, and existing descriptive definitions often implicitly focus on perception-related scenarios. Initially, numerous descriptive definitions of corner cases are proposed. For example, Bolte et al. [26] defined a corner case as *a non-predictable relevant object/class in a relevant location*. Breitenstein et al. [27] emphasized that *a known object in an unusual location* is a corner case. For example, in a traffic scene, if the relationships among participants do not conform to prior knowledge or common sense, the object that violates the rules is defined as a corner case. Li et al. [28] considered the spatial relationship of objects and deemed objects that are about to block a potential path of the self-driving vehicle as the corner case. However, these descriptive definitions of CCs are task-specific. In general, outliers, anomalies, OOD samples, adversary samples, and corner cases are used interchangeably in perception-related research. For example, Heidecker et al. [29] associated corner cases with outliers, novelty, and anomaly samples, and defined sensor-driven corner cases for LiDAR, RADAR, and cameras. In contrast, numerous definitions of corner cases have been developed based on the complexity of the anomalies [27]. The diversity of what may constitute the *corner case* has, to some extent, shaped the solutions to detect these samples. On the other hand, OOD samples are treated as corner cases to assess the trained perception models' generalization to unseen or novel objects. Zhou et al. [30] provided systematic overviews of the detection of out-of-distribution samples. In addition, adversarial samples that can mislead the perception model to induce incorrect decisions are also treated as corner cases [11], [31], [32].

B. Our Contributions

There are numerous survey studies associated with anomaly examples [33], [34], out-of-distribution samples [30], [35], adversary samples [36], policy learning [18], simulators and virtual testing [20], [22], traffic simulation [24], and foundation-model-based scenario generation and analysis [25]. None of these works, however, takes perception-stage corner cases as their central research object, nor do they offer a unified discussion of how such cases ought to be defined, detected, generated, and evaluated—alongside the supporting role of dedicated datasets and simulators—in the context of autonomous driving.

Unlike previous studies, this survey provides a comprehensive examination of perception-oriented corner cases that are crucial for maintaining the operational reliability of autonomous driving systems. As shown in Fig. 1, the integrated use of heterogeneous sensors (cameras, LiDARs, and RADARs) establishes the core data basis for detecting and analyzing these critical scenarios. Despite this technological foundation and recent progress in sensing and anomaly-aware perception that have advanced real-time corner-case detection, there is still no dedicated, comprehensive survey that systematically organizes research on perception-stage corner cases within a unified framework. This study addresses this

gap by offering the first unified survey and taxonomy, thereby clarifying the scope of existing research, improving the effectiveness of anomaly management strategies, identifying open research challenges, and shedding light on their implications for ensuring the safety of autonomous driving. The principal contributions of this work are as follows.

- We conduct a systematical literature overview of studies associated with perception-stage corner cases and provide a multigranularity taxonomy of the existing solutions for the detection of these corner cases. Specifically, we provide the problem definitions of corner cases from three levels, including the data level, the model level, and the semantic level, and formulate these problems mathematically. Meanwhile, a comprehensive literature overview is presented to investigate the existing detection solutions at these three levels in the context of autonomous driving perception.
- As data collection from the real world is time-consuming and expensive, synthesizing large-scale corner cases for model training is a promising solution. In this paper, we also provide an overview of generating corner cases from scene simulators and generation algorithms. In contrast to prior simulator- or scenario-centric reviews [20], [22], [24], [25], our discussion is explicitly organized around how these tools and methods facilitate the study of perception-stage corner cases. In addition, we outline the challenges and research opportunities.

As illustrated in Fig. 2, the remainder of this paper is organized as follows. Section II refines the definitions of perception-stage CCs from three perspectives related to sensors and perception tasks. Sections III and IV then summarize the main approaches for detecting and generating perception-oriented CCs, respectively. Sections V and VI introduce commonly used performance evaluation metrics, large-scale public datasets, and non-commercial simulators tailored to perception-stage corner cases in autonomous driving. Section VII discusses current limitations and outlines challenges for future work on perception-related corner case research. Finally, Section VIII concludes the paper by summarizing key contributions and highlighting future research directions.

II. DEFINITIONS OF CORNER CASES

Understanding corner cases is central to the safety assurance of autonomous driving systems, yet a unified and rigorous definition remains absent. To fill this gap, we propose a **three-level framework** grounded in the end-to-end perception pipeline that spans input, processing, and output. Specifically, the framework is structured along three orthogonal dimensions: **data level**, **model level**, and **semantic level**. Together, these dimensions provide a complete and non-overlapping characterization of CCs. The data level captures abnormalities in raw sensory input, including noise, corruption, and distributional shifts. The model level concerns failures arising from the processing stage, such as limited robustness and unreliable algorithmic behavior. The semantic level addresses deviations in scene meaning, contextual consistency, and safety-critical

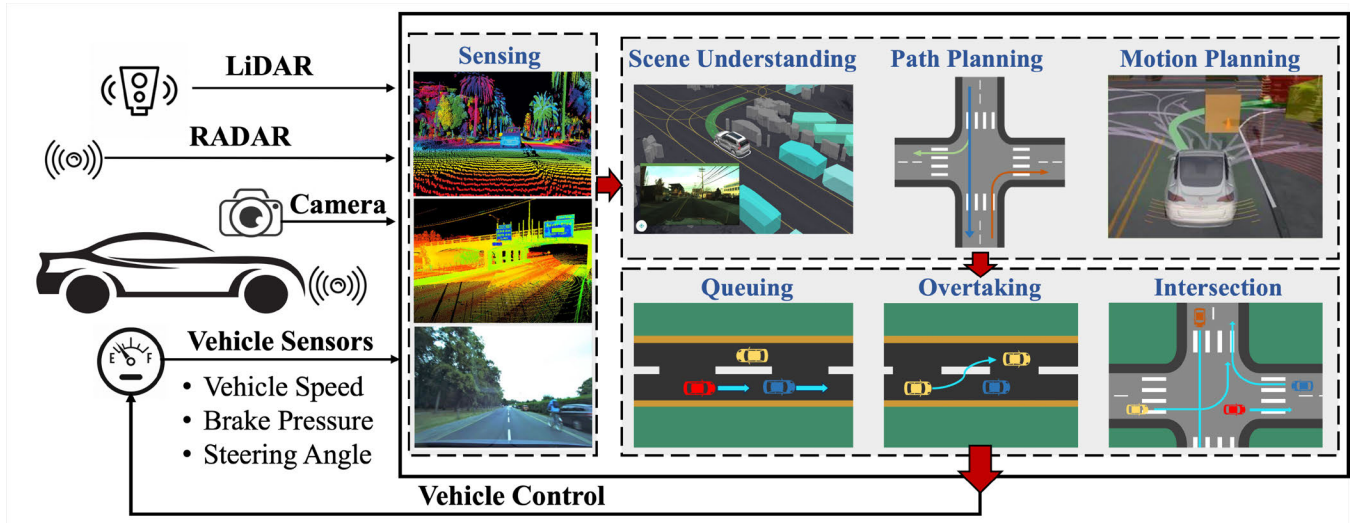


Fig. 1. An overview of the autonomous driving system (ADS). In ADS, a wide variety of sensors, including LiDAR, RADAR, and vision cameras, are integrated for perception tasks such as semantic segmentation, object detection, and instance segmentation. In the decision-making component, numerous automated driving algorithms are employed for surrounding environmental sensing, motion planning, and vehicle control.

interpretation. Since each dimension corresponds to a distinct aspect of the perception chain, the proposed framework ensures both comprehensive coverage and conceptual separation. For cases spanning multiple levels, classification is assigned based on the dominant factor in terms of safety impact. This framework also forms the basis for the CC detection methods introduced in Section III, where the analysis is organized according to the formalization of each level. In the following, we detail the conceptual and mathematical formulations of each dimension and further demonstrate the completeness of the proposed framework.

A. Definition 1: Data-Level Corner Case

Data-level corner cases (CCs) in autonomous driving systems target input-layer anomalies, specifically defined by statistical deviations in raw sensor data that violate pre-established normal measurement ranges—these deviations directly induce inference failures in deployed machine learning models. Notably, such anomalies arise independently of model processing logic or scene semantics; their root causes instead lie in sensor hardware defects or environmental interference. For instance, a camera capturing an image with 90% of pixels saturated to the maximum intensity value due to overexposure, or a LiDAR unit generating point cloud data exceeding its rated maximum effective detection range, both qualify as data-level CCs. In such scenarios, the machine learning model fails to process the input data accurately, primarily due to the novelty and anomalous nature of the sensor measurements.

Mathematically, let the input raw sensor data be denoted as $\mathbf{X} \in \mathbb{R}^n$, where the dimension n is determined by the sensor type: for example, $n = H * W * 3$ for RGB images (with H and W representing the image height and width, respectively), and $n = N * 3$ for LiDAR point clouds (with N denoting the number of points in the point cloud). The detection of data-level CCs is formulated as a binary classification task, with the

objective of determining whether a given input sample constitutes a data-level CC. Formally, we aim to learn a mapping function $y = f(\mathbf{x})$, where the output label $y \in \mathbf{Y} = \{0, 1\}$ —here, $y = 1$ indicates that the input x is a data-level CC, while $y = 0$ indicates normal sensor data. A critical criterion for this classification is: $y = 1$ if and only if the statistical features of x deviate from a pre-defined normal data distribution P_{normal} . This deviation is quantified using a statistical distance metric, expressed as $\mathcal{D}(\mathbf{x}, P_{normal}) > \tau$, where $\mathcal{D}(\cdot)$ denotes a statistical distance metric (e.g., Euclidean distance for evaluating pixel intensity deviations, Mahalanobis distance for measuring LiDAR point coordinate anomalies relative to P_{normal}), and τ represents a pre-determined deviation threshold. For practical clarity, consider 8-bit grayscale images: P_{normal} encompasses pixel intensities within the valid range $[0, 255]$; if x contains any pixels with intensity values outside this range (i.e., > 255 or < 0), the distance $\mathcal{D}(\mathbf{x}, P_{normal})$ will exceed τ , leading to the classification result $y = 1$.

This definition aligns with the sensor-driven CC categorization proposed in [29], which explicitly confirms that input-layer data anomalies are independent of model-specific behaviors and high-level scene semantics—thus solidifying the role of data-level CCs as the foundational layer of our tri-level CC definition framework.

B. Definition 2: Model-Level Corner Case

Model-level corner cases (CCs) capture failures at the processing stage of autonomous driving perception models. They refer to scenarios in which the input data remain statistically normal, i.e., free of data-level anomalies, yet the model nonetheless fails to generate accurate outputs due to deficiencies in its internal representations or decision-making mechanisms. This definition characterizes model robustness along three key dimensions: sensitivity to perturbation, consistency across domains, and completeness of topological representation. In contrast to data-level CCs, such failures

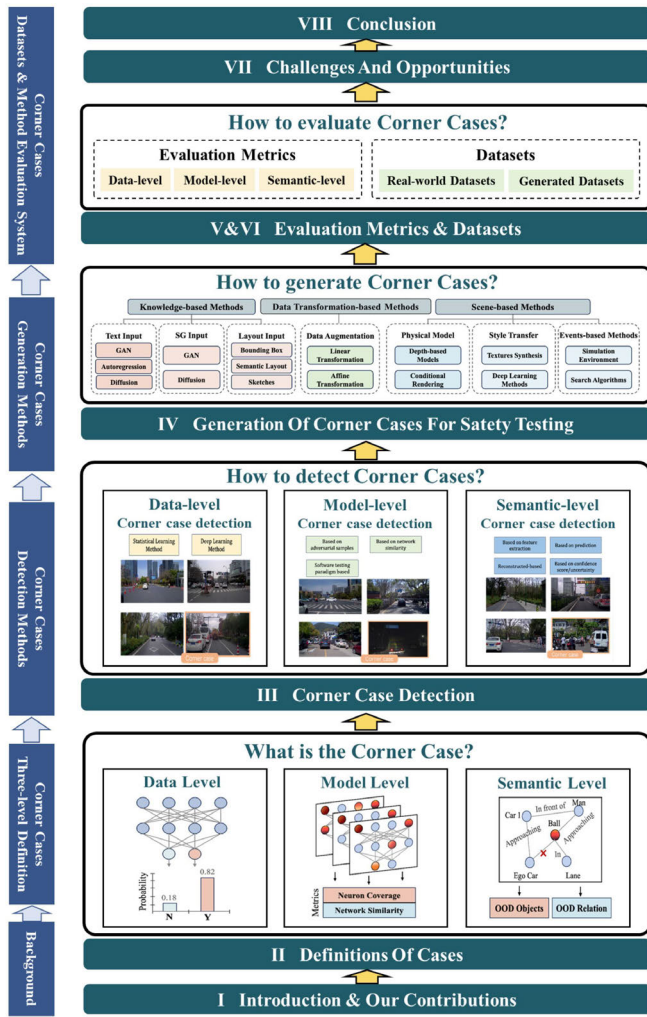


Fig. 2. Overall organization of this paper, including the refined definition of perception-stage corner cases, methods for their detection and generation, evaluation metrics, public datasets, simulation platforms, current limitations, and future research challenges.

cannot be attributed to degraded sensor quality or high-level semantic complexity. Instead, they arise from intrinsic limitations of the model itself, primarily including vulnerability to adversarial perturbations, inadequate generalization under domain shift, and defects in topological structure. The validity of this formulation is further supported by the domain generalization theory established in [30].

Concretely, adversarial perturbations denote small, human-imperceptible modifications to the input that can mislead the model while leaving the underlying semantic content unchanged. For example, a visually normal stop sign with a small adversarial sticker may be incorrectly recognized as a speed limit sign. Mathematically, let $f : X \rightarrow Y$ denote the target perceptual model, $x \in X$ represent normal input data, and δ denote a perturbation vector with $\|\delta\|_p < \epsilon$ (ϵ = perceptibility threshold ensuring δ is human-undetected); an adversarial example qualifies as a model-level CC if $f(x + \delta) \neq f(x)$.

Domain shift-induced CCs occur when test data distribution deviates significantly from training data—for example, a clear-weather-trained semantic segmentation model failing at object boundary detection in fog or heavy snow. Formally, let P_{train}

and P_{test} denote training/test distributions; if the Wasserstein distance (a standard distributional divergence metric) between P_{train} and P_{test} exceeds threshold γ (i.e., Wasserstein distance $(P_{train}, P_{test}) > \gamma$), any $x_{test} \in P_{test}$ is a model-level CC [30]. Topological defects-related CCs, by contrast, stem from insufficient model neuron activation during inference: let $\mathcal{D}$ = all neuron units, $g(x, \mathcal{D})$ = neuron coverage (activated neurons/| $\mathcal{D}$ |, with “activation” = output > 0.1); if $g(x, \mathcal{D}) < \delta$ (δ = minimum coverage threshold from normal dataset validation), x is a model-level CC.

Collectively, a model-level CC is formally defined by two conditions: (1) the model’s output deviates from ground truth ($f(x) \neq y^*$, where $y^* \in Y$ is the true label for x), and (2) the input has no data-level anomalies ($\nexists$ data-level anomaly in x). This dual-condition framework ensures model-level CCs are orthogonal to both data-level and semantic-level CCs in our tri-level system.

C. Definition 3: Semantic-Level Corner Case

Semantic-level corner cases (CCs) refer to failures in scene understanding of autonomous driving systems. They describe scenarios in which the input is statistically normal, with no data-level anomaly, and the model processes the data correctly, with no model-level failure, yet the resulting semantic interpretation of the scene conflicts with driving common sense or prior knowledge. Unlike lower-level CCs, these cases directly affect safety-critical decisions such as trajectory planning and emergency braking and therefore represent the highest level of safety risk, consistent with the observation in [26]. In general, they arise in two forms: out-of-distribution (OOD) objects and OOD spatial relations, both of which violate the expected semantic structure of typical driving scenes.

Concretely, OOD objects are novel or otherwise unusual objects not seen during model training, such as a pet dog suddenly appearing in front of the ego vehicle, an overturned truck blocking a lane, a pedestrian dressed as a doll in the road, or a deer straying onto a highway. OOD spatial relations, by contrast, involve illogical positional relationships between objects that contradict common sense or prior driving knowledge—examples include a chair falling from a moving truck, a phantom pedestrian (e.g., a misdetected or unrealistic pedestrian) “passing through” the middle of the ego vehicle, or a pedestrian walking outside sidewalks and directly crossing the vehicle’s trajectory. These scenarios pose inherent safety risks because they disrupt the model’s expected understanding of reasonable scene dynamics.

Mathematically, let the scene semantic representation generated by the perceptual model be $S(x) = \{O, R\}$, where $O = \{o_1, o_2, \dots, o_k\}$ denotes the set of detected objects in the scene, and $R = \{r(o_i, o_j) | 1 \leq i, j \leq k\}$ denotes the set of spatial relations between objects (e.g., “in front of the ego vehicle”, “crossing the trajectory of”). A semantic-level CC is formally defined by two conditions: ($S(x) \notin \mathcal{S}_{common}$) and there is no data/model-level anomaly in x . Here, $\mathcal{S}_{common}$ represents the set of semantically reasonable driving scenes (e.g., “pedestrians on sidewalks”, “cars traveling in lanes”, “traffic signs above intersections”). For its two specific forms: 1) **OOD objects**: There exists at least one object $o \in O$ such that $o \notin O_{train}$,

where O_{train} is the set of all objects the model encountered during training; 2) **OOD relations**: There exists at least one spatial relation $r(o_i, o_j) \in R$ such that $r(o_i, o_j) \notin R_{\text{common}}$, where R_{common} is the set of logically reasonable spatial relations in driving scenarios. This definition aligns with [26]’s emphasis on semantic-level CCs as the top of the CC hierarchy—their direct link to collision avoidance and driving safety makes them the most critical category in our tri-level framework.

D. Handling Multi-level Corner Cases

In practical autonomous driving environments, corner cases often manifest with hierarchical overlaps, where a single scenario may concurrently exhibit multiple levels of anomalies. For example, an overexposed sensor capture constitutes a data-level anomaly, while an unexpected pedestrian crossing event occurring in the same scene reflects a semantic-level deviation. To systematically address such compound anomalies, we formalize a prioritization framework informed by the scene safety axioms established in [26]. This hierarchical resolution model adheres to a strict precedence order: semantic-level > model-level > data-level, which is derived from the direct impact on operational safety. Semantic-level corner cases pose immediate existential risks to vehicular autonomy, as they disrupt critical path-planning algorithms by introducing contradictory scene interpretations. Model-level anomalies, while less immediately hazardous, undermine system integrity by causing algorithmic failures that lead to misclassification or missed detections. Data-level irregularities, by contrast, represent lower-order disturbances that can be mitigated via preprocessing techniques, such as noise filtering or sensor calibration.

We define categorical decision rules consistent with the following schema: any scenario that exhibits semantic-level discrepancies—regardless of data quality or model performance—is labeled a semantic-level corner case, for example, an overexposed image in which a cyclist unexpectedly appears. If no semantic anomaly is present but the model demonstrably fails on valid input data, the case is classified as model-level, such as misidentifying a traffic cone as a road marker in a clear camera feed. Finally, scenarios that manifest only data-level variations, without degrading model behavior or semantic correctness, are designated as data-level corner cases, such as an overexposed image of an otherwise unobstructed highway. This tri-level (data–model–semantic) framework exhaustively characterizes all CCs in autonomous driving perception systems. Data-level CCs address raw sensor irregularities, consistent with the taxonomy of [29]; model-level CCs capture model failures such as adversarial perturbations, supported by the theory in [30]; and semantic-level CCs encode scene-level safety conflicts, in line with the framework of [26]. Because all CCs arise from these three system components, no additional levels are required. For example, a LiDAR malfunction that triggers both misdetection and misplanning is still a data-level CC, underscoring the framework’s completeness.

III. CORNER CASE DETECTION

We characterize existing solutions for corner-case detection at the data, model, and semantic levels, respectively.

A. Detecting Data-level Corner Cases

In general, existing solutions for detecting data-level corner cases can be categorized into the following two groups.

1) *Statistical Learning Solutions*: A plethora of studies have applied statistical learning algorithms to recognize corner cases at the data level. For example, Liu et al. [37] extracted multidimensional features (e.g., weather, time, location, speed, and surrounding vehicles) from videos recorded by the front-facing camera and applied a random forest algorithm to classify the complexity of driving scenes in a hierarchical manner. Ryan et al. [38] used Gaussian process regression to quantify the uncertainty (confidence interval) of the model prediction. Measurement of risky scenarios in autonomous driving can also be achieved by comparing autonomous and safe driving behaviors [38]. Due to the heavy-tailed distribution of corner cases, the aforementioned solutions exhibit biased performance. To address this problem, a one-class support vector machine (OCSVM) is employed to detect corner cases. The main idea of OCSVM is to map the data into a high-dimensional feature space induced by a kernel function, construct a hyperplane that separates the data from the origin, and maximize the margin between the hyperplane and the origin. This yields a binary function that determines whether a new data point is similar to the training data. The optimization objective function of OCSVM is shown in Eq. 1, where w is the normal vector of the hyperplane, ρ is the distance from the hyperplane to the origin, ζ_i is the slack variable, and $\nu \in (0, 1)$ controls the proportion of outliers and the proportion of support vectors. $\phi(x_i)$ is the kernel function.

$$\begin{aligned} \min_{w, \zeta_i, \rho} & \frac{1}{2} \|w\|^2 + \frac{1}{\nu n} \sum_{i=1}^n \zeta_i - \rho \\ \text{s.t. } & (w^T \phi(x_i)) \geq \rho - \zeta_i, \quad i = 1, \dots, n \\ & \zeta_i \geq 0 \end{aligned} \quad (1)$$

Unlike in classification tasks, corner cases are often treated as anomalies. Therefore, clustering algorithms are widely used to detect data-level corner cases, offering a more intuitive approach: they group samples into homogeneous categories and flag outliers that deviate from the cluster centers. Given the input samples $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$, where $\mathbf{x} \in \mathbb{R}^n$, randomly select the centroid of each category as $\mu_1, \mu_2, \dots, \mu_k \in \mathbb{R}^n$. For each sample $\mathbf{x}_i$, it should belong to $c = \text{argmin} \|\mathbf{x}_i - \mu_c\|_2$. As the sample size increases, the centers of the sample categories tend to remain stable. For example, Lee et al. [39] estimated the cluster centers and covariances of each class and used the Mahalanobis distance to detect OOD samples, while Deng et al. [40] used relative feature displacement to describe the sample as the feature centroids and displacement.

Leino et al. [41] introduced the concept of global robustness of the classification model, and the classifier keeps samples of different categories in a certain width interval in the feature space to ensure the difference between categories. This method leverages the Lipschitz condition to measure neural network robustness. However, such methods perform reliably only on samples with minor perturbations and fail to generalize to data from open environments. Furthermore, samples far from the

centers of their class labels' clusters tend to be misclassified [38], [39], [42].

2) *Deep Learning Methods*: Over the past two decades, deep neural networks have made substantial progress in autonomous driving. For corner case detection, domain adaptation methods are useful for assessing the normality of samples that remain semantically similar to the source domain but differ in distribution. Such domain shifts arise from changes in statistical properties across training, validation, and test data and are commonly grouped into three types: distribution shift, covariate shift, and label shift. "Distribution shift" refers to differences in overall data distributions, "covariate shift" to changes in input features, and "label shift" to changes in label distributions across domains.

However, most existing solutions primarily focus on mapping from a single source domain to a single target domain, resulting in poor generalization. To enhance robustness across multiple source and target domains, a novel Open-Scenario Domain Adaptive Object Detection (OSDA) [43] has been proposed. It utilizes a contrastive vision-language pretraining strategy to distinguish foreground from background and introduces a cross-reconstruction framework across multiple domains to learn domain invariances. Kim et al. [44] extended the domain adaptation theory to LiDAR sensors to enhance performance. They introduced sparsity-invariant feature consistency and semantic correlation consistency to learn generalizable LiDAR representations across the source domain and unseen domains. Moreover, real-world datasets often exhibit class imbalance with a long-tailed distribution. To detect corner cases from unbalanced datasets, Liu et al. [45] proposed OLTR to associate head categories with tail categories by mapping images in space. To address the limitations of the aforementioned methods in handling large-scale, high-resolution image data, Hendrycks et al. [46] introduced the Maxlogit detector to improve performance on multi-label datasets in complex scenes and on anomaly segmentation tasks in driving environments.

Most existing corner-case detection techniques at the data level require the incorporation of additional corner-case datasets to regularize the decision boundaries of the model. Because these auxiliary datasets typically follow a probability distribution that differs from that of the primary training data, the resulting classification boundaries are often insufficiently strict or poorly calibrated. Moreover, the applicability of such methods is generally restricted to relatively simple scenarios, and they do not directly generalize to complex, real-world environments.

B. Detecting Model-Level Corner Cases

To detect model-level corner cases, numerous studies investigated the effects of the input on topological patterns and model behavior.

1) *Solutions Based on Software Testing*: In software development, rigorous testing is paramount for ensuring robustness. This principle has been applied to evaluate the robustness of deep neural networks. Inspired by the concept of code coverage (i.e., the ratio of code statements executed by a test case to all statements), DeepXplore [47] introduced neuron

coverage to measure the proportion of activated neurons by test input. This approach leverages multiple DL models as cross-referencing oracles to identify test inputs that trigger differential behavior and achieve high neuron coverage. Following this paradigm, DeepTest [48], an automated testing framework for autonomous cars driven by DNNs, uses generated test samples across diverse weather conditions to identify extreme behaviors produced by DNNs in extreme scenarios by maximizing neuron coverage. On the contrary, Wu et al. [49] posited that a meaningful corner case detector should be scene-agnostic, inferring that the corner case should be beyond the effective input domain range of the DNN model. Based on this premise, Deep Validation was proposed to evaluate test samples that could cause system errors by examining whether the model's intermediate states exceed predefined thresholds.

However, these solutions are computationally intensive. To mitigate this problem, DeepEvolution [50] proposed a search-based solution that significantly enhances the diversity and neuron coverage of generated test cases.

2) *Solutions Based on Network Similarity*: Compared with neuron coverage, we can extract sample representations from the model's middle layer in a more fine-grained manner to explore corner cases. Dauphin et al. [51] observed that training the same neural network with different random initializations might yield similar performance. Given an in-distribution dataset D_{base} and a corner case dataset D_{cc} , for a neural network $f(\cdot)$, $f_i(\cdot)$ represents the i -th layer of the neural network. If $\text{Similarity}(f_i(x), f_i(y)) < \epsilon$, the input sample is taken as a corner case, where $x \in D_{base}$, $y \in D_{cc}$, and $D_{base} \cap D_{cc} = \emptyset$.

To quantify the similarity of latent representations in neural networks, Wang et al. [52] characterized the structure of matches between neuron activation subspaces. Furthermore, canonical correlation analysis (CCA) [53] is used to quantify the association among multiple high-dimensional variables. Raghu et al. [54] proposed the SVCCA, combining singular value decomposition and CCA, to efficiently compare invariant neural representations and affine transformations. To extend this solution to high-dimensional neural representations, centered kernel alignment (CKA) [55] was proposed to measure the similarity between neural representations derived from different initialization weights.

Inspired by CKA, [56], [57] measured the functional similarity of different models through model stitching. These works primarily examine whether the network can replace the representations of other networks via an affine transformation while preserving functional integrity.

Similar to neuron coverage, Ouyang et al. [58] proposed *surprise sufficiency* to formally quantify how deep learning (DL) systems respond to given inputs. Intuitively, the more surprising an input is to the test DNN, the more likely the system is to behave unexpectedly. First, the activation trace is defined as the pattern of neuron activations triggered by data in the DL system. For a neuron N in the DL system, the activation trace $a_N(x)$ of a given test input x with respect to N is defined as in Eq. 2, where $\alpha_i(x)$ denotes the activation value of x for an individual neuron n_i . Accordingly, the surprise adequacy (SA) for testing a DL system can be formulated based on the

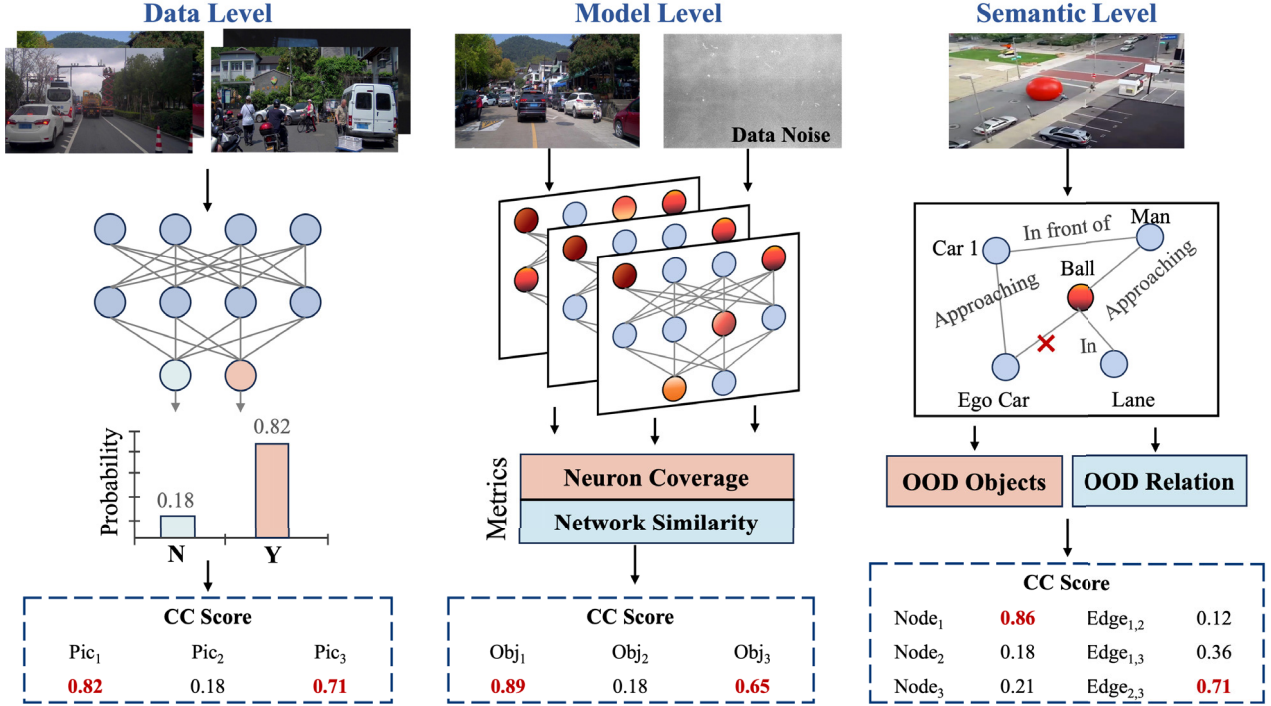


Fig. 3. Understanding corner case from data level, model level and semantic level, respectively: (a) For the data-level CCs, they can be formulated as a binary classification problem; (b) for the model-level CCs, they aim to investigate the model robustness by quantifying the network behaviors under the given samples; and (c) for semantic-level CCs, they focus on unseen objects or scenarios where the spatial relation of objects is rare.

similarity of activation traces between samples.

$$\alpha_N(x) = [\alpha_1(x), \alpha_2(x), \dots, \alpha_N(x)]^T \quad (2)$$

In practical scenarios, there exist outlier instances in which the test and training data are highly similar but differ in their labels. In response to this phenomenon, Ouyang et al. [59] proposed DSA to measure the discrepancy between a single sample and the local descriptor of its class according to Eq. 3, where α_N denotes the activation value (Eq. 2); x_a represents the nearest neighbor in $class_x$; and then, taking x_a as the reference point and finding its nearest neighbor x_b in a class different from $class_{x_a}$.

$$DSA(x) = \frac{\|\alpha_N(x) - \alpha_N(x_a)\|}{\|\alpha_N(x_a) - \alpha_N(x_b)\|} \quad (3)$$

Model-level corner-case detection offers a distinct way to quantify corner cases. Inspired by software testing methodologies, these approaches analyze variations in samples at intermediate network layers using neuron coverage to assess whether a given input elicits abnormal neuron behavior. Unlike other post-processing algorithms, model-level methods are inherently grounded in the model's internal representations, enabling the development of novel strategies for measuring corner cases. However, these methods have poor interpretability, and are difficult to be applied to large-scale datasets directly.

C. Detecting Semantic-level Corner Cases

Unlike the data-level and model-level ones, CCs at the semantic level focus on detecting unseen objects and understanding the abnormal spatial relations of objects. To detect

semantic-level CCs, as shown in Fig. 4, four types of methodologies are studied.

1) *Uncertainty-Aware Solutions*: For uncertainty-aware solutions, they first employ an object detection or segmentation network $g(\cdot)$ to obtain the set of objects $\{obj_1, obj_2, \dots, obj_n\}$ or region proposals in the scene. Then, the confidence score for each object or proposal is quantified using an uncertainty estimator. Finally, objects with higher uncertainty are recognized as corner cases. As shown in Fig. 4, numerous studies followed this paradigm to produce confidence scores or uncertainty scores for objects, regions, or instances [60], [61], [62]. This method is simple and effective, without modifying the structure of the classification model or training an OOD sample classifier.

Similar to the confidence score, numerous studies [63], [64], [65] have detected abnormal regions in images by outputting their uncertainty. DUQ [63] used a deterministic deep neural network for uncertainty estimation, using a trainable kernel function to reject data points outside the distribution. Kendall et al. [64] used Monte Carlo sampling and Bayesian inference to estimate the uncertainty of each pixel. However, none of them can guarantee real-time uncertainty prediction. Huang et al. [65] attempted to use video sequences as input and leverage temporal information to aggregate and disperse the uncertainty of each pixel. In general, video sequences contain more semantic information and are more consistent with data from autonomous vehicle sensors. Joseph et al. [66] performed contrastive clustering on features generated by the residual module in the ROI head, and used energy-based

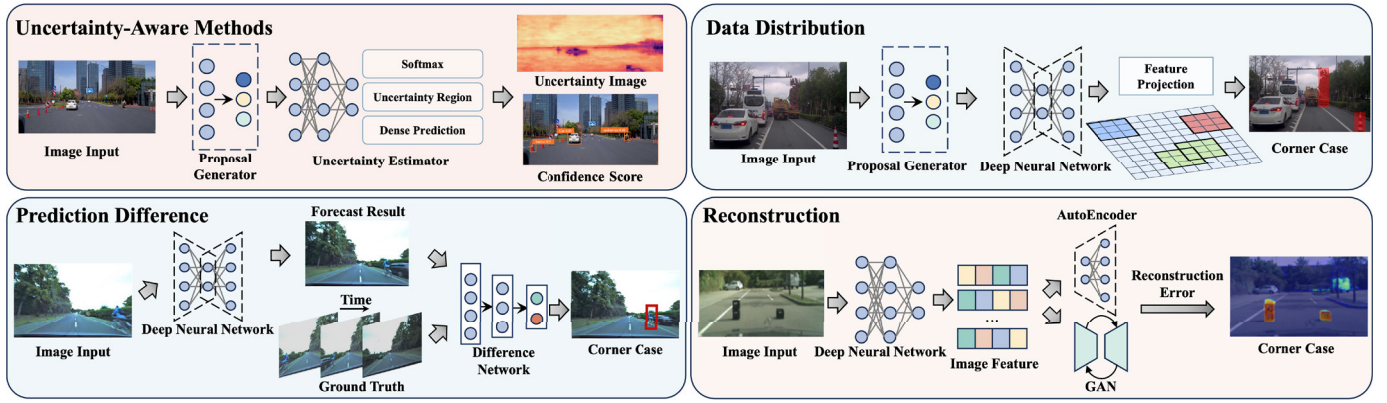


Fig. 4. An overview of detecting semantic-level corner cases. For corner-case detection at the semantic level, we divide it into four types in the figure: Uncertainty-Aware Methods, Data Distribution Methods, Prediction-Difference-based Methods, and Reconstruction-Based Methods. We summarize four methods that prior research collectively regard as the paradigm.

models to learn experience categories and identify possible location types in the test set.

To encapsulate such methodologies, uncertainty and confidence measures are typically implemented as post-processing. This can include either outputting the softmax posterior probability or computing the distance between the sample and the prototype. In essence, most methods are evaluated by comparing the score to a fixed threshold to determine whether the sample is a corner case. Unlike data-level detection methods, this type of method generally does not require additional auxiliary data. Instead, it conducts supervised training directly through standard training samples.

2) *Solutions Based on Prediction Difference*: Prediction-based methods typically utilize video sequences as input to the model, which can be formulated as $\mathbf{d}_{t+1} = s_{t+1} - f(s_t)$, where $\mathbf{d}_{t+1}$ is the prediction error, s_{t+1} is the ground truth, s_t is the input scene and $f(\cdot)$ is a complex-valued function that predicts the next scene. A salient characteristic of corner cases is that they are difficult for general detection models to recognize and process correctly. By comparing the difference between the predicted scene s'_{t+1} and the ground truth s_{t+1} , the detection of the corner case object in the scene is achieved. Therefore, unlike the aforementioned methods based on confidence scores, the difference between the prediction result and the ground truth is used to locate the object in the corner case [26], [28], [67]. As illustrated in Figure 4, the method predicated on the prediction difference diverges from other methods that use a single-frame image as input. It requires a video sequence as ground truth to determine whether the model has accurately discerned the object's intention in the scene at the subsequent moment.

Bolte et al. [26] designed an image prediction module that computes the prediction error for each input frame, performs semantic segmentation on the input frames to identify and locate objects in the scene, and devises a detection system that fuses the information from image prediction and semantic segmentation. This framework verified whether the predictable objects/classes/instances are in the expected locations and output a binarized score of the corner-case probability. Compared to the method proposed in [26], Liu et al. [67] use GAN to

assist in the prediction of new frames. Through adversarial training of GAN, to ensure the accuracy and authenticity of the prediction, and then improve the performance of frame prediction. In [28], multiple modalities, including point clouds and images, were utilized to bolster corner case detection robustness. In autonomous driving scenarios, the vehicle's sensors are typically multimodal, including cameras, radars, and lidars. In addition to using single-modal camera data for corner case detection, Bogdoll et al. [68] proposed a method to detect anomaly data based on multimodal data.

3) *Reconstructed-Based Solutions*: The underlying assumption of reconstruction-based solutions is that normal scenes tend to have smaller reconstruction errors than corner cases. By comparing the reconstructed image with the original image, the corner case can be identified and localized. As shown in Fig. 4, it first feeds the raw image into a neural network to learn the intermediate representations. Then, a reconstruction task is introduced to generate images based on the intermediate representations. If the discrepancy between the reconstructed output and the original input exceeds a certain threshold, the object can be classified as a corner case.

In general, autoencoders are widely used for a range of reconstruction tasks [69], [70]. For example, Hasan et al. [69] used autoencoders to capture the distributions of normal motion patterns from video sequences and detect abnormal behaviors based on reconstruction errors. Xia et al. [71] introduced a discriminative autoencoder that can identify abnormal images by incorporating discriminative information during training. However, these methods are not feasible for corner cases that are rare or unseen. For the unseen samples, [70] developed a category-conditioned autoencoder that first trains an encoder on a closed set of categories and then trains a decoder conditioned on the category labels to reconstruct the input. The reconstruction error is then used to determine whether the input belongs to a known or unknown category.

On the other hand, semantic and instance segmentation models are also widely used for reconstructive tasks. For semantic segmentation, the goal is to assign each pixel in an image to a class or object label. Accordingly, numerous studies use the output of a semantic segmentation network

as an intermediate representation to reconstruct images and explore corner-case objects in autonomous driving scenes [72], [73], [74]. For example, Yan et al. [75] used a pre-trained image segmentation model to produce a segmentation map, and then used the segmentation masks as input to a conditional GAN for image generation. They then used the dissimilarity network to detect differences between the two images and localize uncertain objects in the scene. To enhance robustness, Di Biase et al. [73] combined multiple information sources, including semantic maps, softMax entropy, softMax distance, regenerated images, and perceptual differences, into a difference network to recognize abnormal objects in images. In addition, depth information can play an auxiliary role in corner-case detection. Furthermore, Lis et al. [76] detected and identified road obstacles by erasing instance objects. The model first uses a general method for filling, relying on adversarial methods to ensure that the erased instances in the filled image are partially realistic. Secondly, a separate network is trained to compare the two images and detect abnormal road objects. In essence, the reconstruction method is based on learning intermediate representations of images and on the reconstruction discrepancy. This leads to an over-reliance on feature extraction, semantic segmentation, and related techniques. This will directly influence the subsequent reconstruction effect, thereby affecting our judgment of corner-case objects.

4) *Solutions Based on Data Distributions:* Corner cases (CCs) typically follow distributions in latent representation space that differ from those of normal samples. Building on this observation, data distribution-based methods map input images into a shared latent space via learnable functions, where distributional discrepancies are measured using kernel functions or random feature mappings, typically using maximum divergence criteria. In this framework, deep neural networks first extract features from the input, which are then either processed further by downstream methods or used directly for classification. Unlike uncertainty-based approaches, this class of methods either directly identifies sample types or performs post hoc classification based on the extracted latent features. For example, two additional constraints are enforced in DaCUP [77] for representation learning: not belonging to a modelable category and not being visually similar to non-anomalous objects in the image. Based on these assumptions, the authors propose a novel embedding bottleneck to effectively capture the visual variations in unknown, multimodal scenes. Reference [78] integrated both greedy search and confidence intervals to determine the boundaries of abnormal regions in multivariate spatio-temporal time series. Reference [79] presented a CROSR open-set classification framework, which consists of two components: a closed-set classifier and an unknown detector. The closed-set classifier uses common supervised learning methods to learn label representations. The unknown detector uses the reconstructed potential representations and labeled data for training.

Semantic-level corner-case detection allows us to pinpoint specific objects or instances within an image, thereby improving both interpretability and practical utility. By operating at

this finer granularity, we can focus on specific regions or objects exhibiting unusual behavior, rather than treating the entire image as an outlier. Addressing corner cases at the semantic level can therefore enhance the robustness and safety of ADS. However, in contrast to image-level classification, which considers the entire scene as a single entity, semantic detection works at the object level, substantially increasing task complexity and demanding more sophisticated models and stronger regularization. Moreover, current datasets primarily focus on common objects and scenes, leading to degraded performance in corner cases.

IV. GENERATION OF CORNER CASES FOR SAFETY TESTING

Autonomous driving systems (ADSs) rely extensively on large-scale, real-world driving datasets for perception, prediction, planning, and validation. However, exclusive dependence on such datasets is both economically prohibitive and intrinsically limited. First, the collection and annotation of autonomous driving data are time-consuming, labor-intensive, and costly [80], [81]. Second, the inherently dynamic nature of environmental and traffic conditions makes exhaustive coverage of all possible scenarios infeasible, which in turn leads to degraded model performance under distributional shifts and weak generalization to previously unseen situations [13], [82]. Third, real-world data predominantly reflect routine operating conditions rather than rare or challenging corner cases, making such data inefficient for safety-critical stress testing. Consequently, corner-case generation should be regarded not merely as a data synthesis task, but as a systematic mechanism for enhancing model adaptiveness and robustness in long-tail and safety-critical scenarios. Existing studies provide preliminary empirical support for this perspective. For instance, Bewley et al. [83] demonstrated that a self-driving vehicle trained exclusively on simulated data could operate on real roads, indicating that synthetic data can effectively support downstream driving performance. More broadly, synthetic data has been shown to accelerate the development of generalizable learning-based algorithms [84]. Therefore, the generation of corner cases is pivotal for the safety testing of ADSs. In this section, we review corner-case generation methods from three perspectives: data transformation-based, knowledge-based, and scene-based (Fig. 5).

A. Data Transformation-Based Corner Case Generation

Image transformation-based corner case (CC) generation typically applies targeted augmentation operators to derive new samples from existing datasets. This approach leverages specialized image processing techniques, including sharpening, rotation, scaling, and noise injection, to generate scenario-specific samples that mimic adverse conditions such as rain, fog, and low light, which are common sources of corner cases in autonomous driving [8], [48]. For example, DeepTest [48] applied affine transformations to original scenes to evaluate ADS robustness under variable weather conditions. DeepXplore [47] generated transformed inputs to maximize neuron coverage, while DeepRoad [85] adopted a metamorphic testing

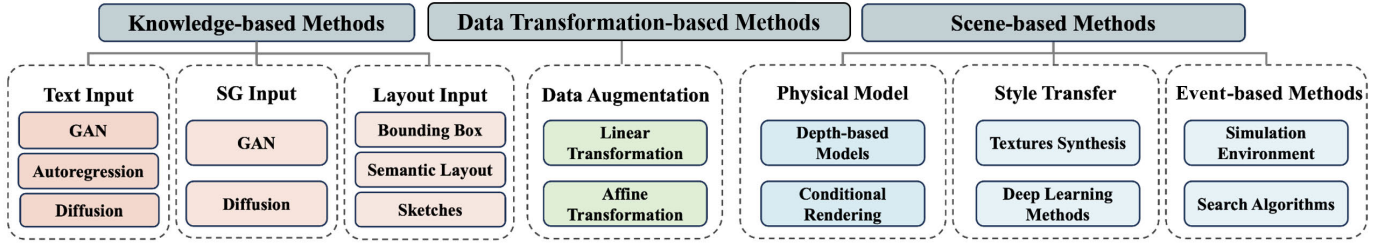


Fig. 5. An overview of corner case generation methods. For data-augmentation-based tasks, transformation operations encompass both linear and affine transformations. Knowledge-based methods can be categorized into three types: text, scene graphs, and layouts. Scene-based generation methods can be categorized into physical models, style transfer, and event-based approaches.

framework to expand test coverage through transformation functions that produced new samples with different characteristics.

A principal advantage of these methods lies in their capacity to enhance the robustness of ADS to nuisance factors and domain shifts encountered in adverse environments. By introducing controlled perturbations while preserving the core scene semantics, they mitigate dependence on spurious correlations typically learned under standard conditions and improve robustness to variations in weather, illumination, and visual appearance. This property is particularly beneficial in scenarios where real-world corner-case data are scarce. Nonetheless, these approaches primarily address appearance-level corner cases, and the generated samples often inherit biases from the original dataset. Furthermore, the perturbations applied are generally restricted to relatively simple or limited scenarios.

B. Knowledge-Based Corner Case Generation

To address the limitation of transformation-based methods—their inability to capture the semantic complexity of real traffic scenes—knowledge-based corner case generation incorporates external knowledge (e.g., text descriptions, scene graphs, layouts) to guide image synthesis with richer structural and semantic constraints. By integrating metadata such as object orientation, size, color, and spatial relations, these methods enable the construction of rare scenarios that are not readily obtainable from routine driving logs.

1) *Text-Guided Generation*: As a natural and flexible form of scene description, text provides an intuitive source of semantic guidance for image generation. Reed et al. [86] proposed the classic model to generate images based on descriptive text input, and StackGAN [87] was applied to improve the image resolution from 64×64 to 256×256 . Since then, numerous GAN-based text-to-image generation models have been proposed [88], [89], [90]. In addition, the attention mechanism was integrated into the GANs to align the same objects in text sequence and visual semantics [91], [92]. Quan et al. [93] combined the attention regularization module and the region proposal network (RPN) to efficiently locate keywords. With the popularity of Transformer [94], autoregressive syntactic models employ Transformer-based sequence-to-sequence architectures to learn the relationship between linguistic input and visual output. Autoregressive models were usually combined with variational autoencoders

(VAEs) [95] and CLIP [96]. For example, Ramesh et al. [97] trained VAE to acquire image tokens. The text and image tokens were concatenated and then fed into the Transformer. Similarly, CogView [98] and UMT (Unifying Multimodal Transformer) [99] conducted large-scale joint generative pre-training on text and image tokens obtained from VQ-VAE.

Recently, denoising diffusion models have gained increasing attention and have demonstrated outstanding performance in image generation [100], [101]. It is a parameterized Markov chain that transforms Gaussian noise into samples. The model is meticulously trained through variational inference, enabling it to generate samples that match the data distribution. The mathematical representation of the forward process (diffusion process) is shown in Eq. 4, where x_t and x_{t-1} are random variables, $\mathcal{N}$ represents the normal distribution, $\sqrt{1-\beta_t}x_{t-1}$ is the mean, $\beta_t I$ is the covariance matrix, and I is the identity matrix. The input to the forward process is the original image, which follows a specific distribution, and the output is a noisy image that follows a Gaussian distribution.

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I)$$

$$q(x_{1:T} | x_0) := \prod_{t=1}^T q(x_t | x_{t-1}) \quad (4)$$

The mathematical representation of the backward process (reverse process) is shown in Eq. 5, where $p_\theta(x_{0:T})$ represents the probability density function of the random variable sequence from x_0 to x_T ; $p_\theta(x_{0:T})$ is the probability density function of x_T ; $\prod_{t=1}^T$ represents the product of t from 1 to T . $p_\theta(x_{t-1} | x_t)$ is the conditional probability density function between x_{t-1} and x_t under given conditions. With a parameter θ , $\mu_\theta(x_t, t)$ is the mean function, and $\Sigma_\theta(x_t, t)$ is the covariance matrix function.

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t)$$

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (5)$$

To reduce training costs and substantial computational load, Rombach et al. [102] introduced the latent diffusion model, which demonstrated robustness in generating high-resolution images, including some corner cases.

Text-guided generation's key advantage is explicit specification of rare semantic scenarios absent from the original dataset: researchers can directly describe such cases in language and

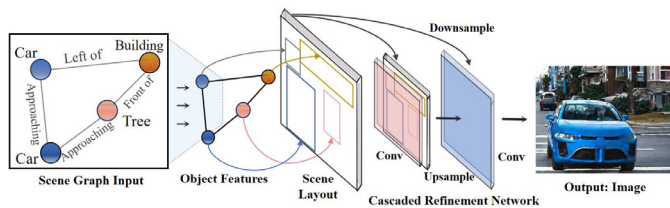


Fig. 6. Overview of Sg2im. The input scene graph specifies objects and their relationships, which are then processed using a graph convolutional network. After computing embedding vectors for all objects, they are used to predict bounding boxes and segmentation masks, then combined to form a scene layout. The layout is converted to an image using a cascaded refinement network (CRN).

synthesize corresponding scenes, rather than waiting for them to emerge naturally in real-world data. This makes it ideal for constructing long-tail scenarios, offering semantic controllability at low data-collection cost.

2) *Scene Graph-Guided Generation*: Scene graph (SG) is a widely used data structure in computer vision for scene understanding. In a scene graph, nodes correspond to objects or entities, while edges represent their relationships or interactions [103]. Mathematically, given a set of objects C_o and a set of relations E , a scene graph is a tuple (O, E) , where $E \subseteq O \times R \times O$ is a set of directed edges of the form (o_i, r, o_j) and $o_i, o_j \in C_o$, $r \in E$. Therefore, precise control over the semantic position of traffic elements plays a vital role in corner case generation. Sg2im [104] was the first to use graph convolution for SG processing, learning layout information by predicting the bounding boxes and segmentation masks of objects (Fig. 6), which the cascade refinement network then converts into images. Following this paradigm, Mittal et al. [105] subsequently proposed a sequential incremental approach. In contrast, Ashual and Wolf [106] distinguished objects by adding attributes such as types and appearance embeddings, employing dual encoding for each object. Attention mechanisms [107] and external knowledge bases [108] have been further employed to enhance the model’s understanding of objects. Scene graphs are useful data structures for describing traffic scenes. To bridge the gap between real-world traffic images and generated images, Savkin et al. [109] proposed a method based on domain-invariant scene representation to directly generate traffic imagery. The model extracted the SG from the virtual environment and then generated traffic scenes in the real environment. Compared with previous methods that relied on manually crafted SG representations, the images produced by this method appeared more realistic and compliant.

SG-guided generation is particularly effective for explicitly encoding inter-object relationships and atypical interactions, which constitute the core of most safety-critical driving scenarios. In contrast to appearance-level augmentation, SG-based approaches enable the systematic construction of corner cases characterized by abnormal object configurations, complex agent interactions, and traffic-rule-related semantic violations. Consequently, these methods increase both visual variability and semantic representativeness, which are crucial for rigorously assessing and improving model performance in interaction-intensive driving environments.

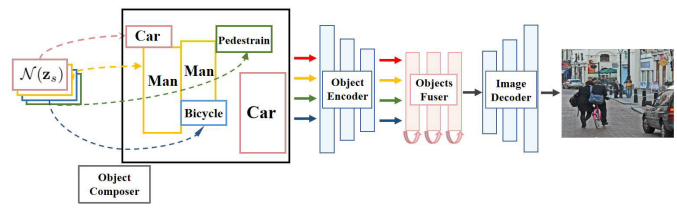


Fig. 7. Image generation from layout. Given the coarse layout (bounding boxes and object categories), the Layout2Im model simplifies the process by sampling each object’s appearance from a normal distribution. They are then transformed into a real image by a series of components. This figure is inspired by [110].

3) *Layout-Guided Generation*: Layout has historically functioned as an intermediate representation for processing textual and other multimodal inputs. In particular, its capacity to encode object locations—an aspect that is crucial in traffic scenes—has motivated researchers to employ layout information directly as an input modality to constrain the spatial distribution of scene objects. Consequently, layout-guided image generation methods leverage explicit positional information of scene elements to condition and structure the synthesis process. In general, layout representations are commonly instantiated as “bounding box + category,” semantic layout maps, and related variants. For example, Zhao et al. [110] proposed a classic layout-to-image model, as shown in Fig. 7, where the input consisted of bounding boxes and object categories. It constructed feature maps for individual objects and applied a convolutional LSTM to generate a hidden feature map. Subsequently, it decoded the entire map into a set of output images. LostGan [111] focused on reconfigurable layout and style, and also learned about object instance-specific layout-aware feature normalization (ISLA) in generators. ISLA was extended to Attr-ISLA with additional control over the attributes in [112]. Furthermore, [113], [114], [115], [116] optimized the architecture to produce images with more robust object appearance in complex layouts. Zheng et al. [117] utilized structural image patches with region information and a diffusion model to design an end-to-end controllable model.

In addition to bounding boxes, the semantic layout offers a flexible description. Park et al. [118] introduced spatially adaptive normalization, which uses the layout to modulate activations in normalization layers. Following this paradigm, Zhu et al. [119] further conditioned on segmentation masks that described semantic regions in the output images. Additionally, [120], [121] used semantic operations to modify local objects in images. The generated traffic scenarios could produce more reasonable and distinctive results for fine-grained objects. Using a multiscale guided diffusion model, Zeng et al. [122] leveraged a precision-encoded mask pyramid and combined text features by accuracy level.

Layout-guided generation is well-suited to synthesizing rare, safety-critical spatial configurations underrepresented in routine driving data (e.g., unusual object placement, constrained road occupancy, anomalous spatial proximity). Layout control enables a more direct specification of such configurations than standard augmentation, providing a practical way to

model atypical spatial arrangements that challenge the ADS perception and planning modules.

C. Scene-based Corner Case Generation

For robustness of the ADS model, the data pipeline must incorporate diverse driving scenes, and scene-based corner-case generation methods address this by moving beyond image-level modifications to directly create safety-relevant scene-level variations—either by generating scenarios with real-world-like distributions or by collecting samples under targeted conditions. These methods are primarily categorized into physical-model-based, style-transfer-based, and event-based approaches.

1) *Scene Generation Based on Physical Models:* Physics-based approaches rely on explicit physical models, including atmospheric scattering models [123], [124], rain photometric modeling [125] and pulse propagation models in the presence of scattering particles [126], [127], [128] to generate samples under foggy, rainy, or snowy conditions [8]. Physics-based generation solutions are efficient and interpretable. However, the model parameters are predefined under ideal assumptions and only apply to specific scenes. For example, in the atmospheric scattering model, the atmosphere condition is assumed to be homogeneous and overlooks the impacts of texture density and scene depth on light [129]. To ensure that the predefined parameters adapt to varying conditions, it is essential to adjust them within a specified range. This enables control over specific attributes of generated scenes. Physical-model-based approaches are particularly useful for generating adverse environmental conditions in a controllable and interpretable manner, as such conditions are a major source of distribution shift for perception systems. Given that weather-related corner cases are costly and challenging to capture exhaustively in real-world driving datasets, physics-based models provide a principled framework for assessing ADS performance under environmental degradation while maintaining precise control over relevant scene parameters and attributes.

2) *Scene Generation Based on Image Style Transfer:* Image style transfer has been applied in autonomous driving to improve robustness to target domains, such as adverse weather conditions and day-to-night image translation [131]. Applying style transfer to existing driving images can streamline corner-case generation and reduce the time and resources required to construct domain-diverse samples.

Traditional style transfer methods rely on manually crafted low-level features to align patches between the content and style images [132], aiming to achieve a harmonious fusion of their visual characteristics. In recent years, deep learning methods have demonstrated strong capabilities for feature extraction, offering a promising approach to capture intricate style patterns [133], [134]. Arbitrary style transfer methods used unified models with feedforward architectures to efficiently handle diverse inputs [135], [136], [137]. Liu et al. [138] used an adaptive attention normalization module to acquire spatial attention scores from both shallow and deep features. However, image reconstruction errors can corrupt image content after several rounds of stylization. An et al. [139] mitigated content leakage employing reversible neural flows and an

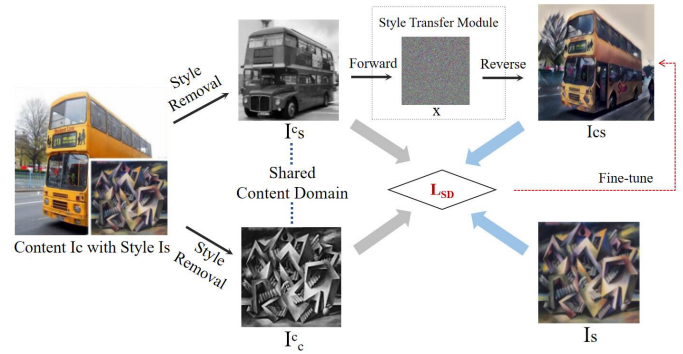


Fig. 8. Overview of StyleDiffusion. Initially, the content image I_c and the style image I_s are processed by a diffusion-based style removal module to extract domain-aligned content information. Subsequently, the content of I_c is utilized in a diffusion-based style transfer module to produce the stylized result I_{cs} . This figure is inspired by [130].

unbiased feature transfer module. It performed unbiased style transfer on deep features and reconstructed the stylized images through reversed feature inference. Zhang et al. [140] applied contrastive learning to directly learn style representations from image features, improving arbitrary style transfer in different domains.

In addition, visual transformers were incorporated into style transfer tasks. Park and Kim [141] performed global style composition guided by content through a transformer-driven style composition module. They used self-attention mechanisms to capture long-range dependencies and subsequently merged Styleformer with StyleGAN2 to generate compositional scenes. The model could utilize the Cityscapes dataset to produce high-resolution samples depicting traffic and vehicle conditions. Reference [142] proposed a transformer-based method to alleviate biased content representation in style transfer by accounting for long-range dependencies of input images. In diffusion-based style transfer, [143] proposed an inversion-based method for stylization. Furthermore, [130] explored the problem of content–style disentanglement, as shown in Fig 8. The diffusion-based style removal and transfer modules first separated style information from both content and style images to reveal domain-aligned content, and then transferred the disentangled style representation to the target image.

Style-transfer-based data generation constitutes an effective strategy for exposing models to cross-domain appearance variations while preserving the underlying scene semantics. This is particularly pertinent in the context of autonomous driving, where an identical physical traffic configuration may exhibit substantial variability across different weather conditions, illumination regimes, or sensing modalities. By augmenting domain diversity around fixed scene semantics, style transfer facilitates model adaptation to target conditions that are infrequently represented or entirely absent in the original source domain.

3) *Scene Generation Based On Events:* Among scene-based approaches, event-based corner-case generation is most directly tied to safety validation, as it focuses on algorithmically identifying safety-critical corner cases (e.g., traffic

TABLE I
SUMMARY OF VIRTUAL ENVIRONMENT SIMULATORS

Simulator	System	Website	OS	Status	Scene and Function
CarSim	Win/Linux	https://www.carsim.com/products/carsim/	No	Active	Single vehicle dynamics simulation and 3D map scene.
Carla [145]	Win/Ubuntu	http://carla.org	Yes	Active	City 3D map scene, supporting various sensors, used for the development of a single-vehicle driving system from perception to control.
MetaDrive [146]	Win/Linux/Mac OS	https://github.com/metadriverse/metadrive	Yes	Maintained	3D road network scene, free to build roads, with packaged RL environment.
AirSim [147]	Win/Linux/Mac OS	https://microsoft.github.io/AirSim/	Yes	Archived	City and field 3D map scenes, supporting various sensors, and focusing on the perception and control of a single vehicle.
TORCS	Win/Linux	http://torcs.sourceforge.net	Yes	Active	Urban and outdoor 3D racing scenes, supporting sensors that contain semantic information, mainly for vehicles aimed at racing.
SUMO [148]	Win/Linux/Mac OS	https://www.eclipse.org/sumo/	Yes	Active	2D road network scene, supporting custom roads and vehicles, traffic lights, etc.
VISTA 2.0 [149]	Win/Linux/Mac OS	https://vista.csail.mit.edu/index	Yes	Maintained	Generated from real scenes and supports multi-sensor perception simulation.

TABLE II
EVALUATION METRICS

Metric	Formula/Description
Precision	$\frac{TP}{TP+FP}$
Recall	$\frac{TP}{TP+FN}$
F1	$2 * \frac{Recall * precision}{Recall + Precision}$
AP	The Area Under Precision-Recall Curve
AUC	The Area Under the ROC Curve
Neuron coverage	$NCov(T, \mathbf{x}) = \frac{ n \forall x \in T, out(n, \mathbf{x}) > t }{ N }$
SSIM	$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$
mIoU	$IoU_t = \frac{TP_t}{TP_t + FP_t + FN_t}$
Wilderness Impact	$Wilderness Impact(WI) = \frac{P_K}{P_{K \cup U}} - 1$

¹ TP : True Positives; TN : True Negatives; FP : False Positives; FN : False Negatives.

violations, collisions) that pose unsafe outcomes prior to real-world deployment [144]. Rather than generating only visually distinct scenes, these methods search the action space to pinpoint behaviors that trigger such safety risks. The sensory outputs associated with these trigger events are then used as corner cases. This formulation makes event-based generation particularly meaningful because it targets the conditions under which model failure is most consequential.

Inspired by parallel system theory [150], as shown in Table I, numerous simulators have been developed, including CARLA [145], VISTA data-driven simulation engine [149], [151], urban mobility simulation (SUMO) [148], InterSim [152], TrafficBots [153], and HMIway-env [154]. Leveraging these simulators, numerous search algorithms are employed to efficiently identify rare safety-critical events in a large search space. Reinforcement learning (RL) and its variants are

widely used to control agents' intentional behavior, simulate scenarios, and identify dangerous actions that violate them. For example, Sun et al. [155] used reinforcement learning to generate corner-case scenarios in the simulator's virtual environment. Kishore et al. [156] first selected failure cases, such as false-positive and false-negative samples, and used imitation training to learn an optimal policy for generating data with domain randomization. Due to the rarity of safety-critical events, Feng et al. [1] proposed a dense deep reinforcement learning approach (D2RL) to find critical states (dangerous events) from a large number of vehicle behavior plans, and reconnect these critical states for information densification through augmented reality to accelerate model training in potentially unsafe situations. To improve the efficiency of model training and robustness under unseen 'long tail' scenarios, Cao et al. [157] presented a dynamic confidence-aware reinforcement learning (DCARL) framework to generate vehicle potential trajectories, evaluate the dynamic confidence value of each trajectory, and update the policy with the high confidence value for continuous performance improvement.

Event-based generation goes beyond improving sample realism; more importantly, it offers a direct mechanism for uncovering failure-inducing states and safety-critical interactions that are difficult to encounter naturally. Compared with other generation strategies, event-based methods are more tightly coupled to ADS failure modes because they explicitly search the regions of the scenario space where robustness is most severely challenged. This makes them particularly valuable for safety-oriented testing, long-tail validation, and the iterative improvement of model behavior under rare but hazardous conditions.

V. EVALUATION METRICS

Well-designed evaluation metrics are essential for determining whether AVs are truly ready for deployment, especially in rare but safety-critical edge cases [1], [157], [158]. In

TABLE III
A COMPARATIVE STUDY OF EXISTING DATASETS FOR AUTONOMOUS DRIVING

	Datasets	Sensors	Scene	Data type	Scale	Task types	Annotated frames for						
							Seg.	2D Det.	3D Det.	MOT	Depth	Flow	Pose
Real-world Data	NuScenes [161]	lidar camera	urban	trajectories HD map	1000 driving scenes	semantic segmentation instance segmentation	-	-	✓	✓	-	-	-
	KITTI [162]	lidar camera	urban highway	image point cloud	50 sequences	depth completion Object detection depth prediction	-	✓	✓	✓	✓	✓	-
	CODA [28]	camera	urban	image	1500 driving scenes	instance segmentation corner case detection	-	✓	-	-	-	-	-
	Waymo Open [163]	lidar camera	urban	trajectories HD maps	1150 scenes, 20s	2D/3D object detection and tracking tasks	-	✓	✓	✓	-	-	-
	A2D2 [164]	lidar camera	urban highway rural	image point cloud	40000+ frames	semantic segmentation instance segmentation 3D tracking	✓	✓	✓	-	-	-	-
	BDD100K [165]	camera	urban highway rural	image video	100,000 HD videos, 40s	semantic segmentation instance segmentation Detection and tracking	✓	✓	-	✓	-	-	-
	CamVid [166]	camera	urban	image video	700+ images	semantic segmentation	✓	-	-	-	-	-	-
	Road Damage [167]	camera	urban	image	31000 images	road damage detection	-	✓	-	-	-	-	-
	RoadAnomaly [72]	camera	urban	image	60+ anomaly images	anomaly detection intent prediction free QA risk detection	-	✓	-	-	-	-	-
	DRAMA [168]	camera	urban	video	17785 scenario clips		-	✓	-	-	-	-	-
Generated Data	Mapillary Vistas [169]	-	urban	image	25000 images	semantic segmentation instance segmentation	✓	✓	-	-	-	-	-
	SYNTHIA [170]	-	urban highway	image	200,000+ images	Image segmentation	✓	✓	✓	-	-	-	-
	VIPER [171]	-	urban	image	600+ pedestrians 1200+ images	Person Re-identification	✓	✓	✓	✓	-	✓	-
	AIOdrive [172]	-	urban	image point cloud	100 sequences with 1000 frames	depth completion semantic segmentation tracking prediction	✓	✓	✓	✓	✓	-	-
	GTA-V [173]	-	urban	image	24000+ images	semantic segmentation	✓	-	-	-	-	-	-
	SHIFT [174]	-	urban rural	image	4800+ sequences	semantic/instance segmentation Object detection MOT trajectory prediction	✓	✓	✓	✓	✓	✓	✓
	MUDA [175]	-	urban	image	10413 images	semantic segmentation depth estimation object/instance detection	✓	✓	-	-	✓	-	-
	Apollo [176]	-	urban indoor highway rural	image	273k scenes	semantic/instance segmentation Object detection	✓	✓	✓	-	-	-	-

this section, we summarize three principal categories of metrics—data-level, model-level, and semantic-level—which collectively constitute a systematic framework to quantify the safety of automated vehicles.

A. Data-Level Metrics

In data-level corner case detection, which aims to identify anomalous inputs not covered by standard training data, most methods treat the task as binary classification by separating rare, safety-critical corner cases from normal samples. The most common evaluation metrics are AUC and AP, which capture complementary aspects of detection performance and together provide a more complete assessment. AUC is a standard metric for evaluating a model's overall discriminative ability. It is computed from the receiver operating characteristic (ROC) curve, which characterizes performance across all possible classification thresholds by plotting the true positive rate (TPR), namely the proportion of correctly identified corner cases, against the false positive rate (FPR), namely the proportion of normal samples incorrectly classified as corner cases. AUC aggregates this tradeoff into a single score: a value of 1 reflects perfect discrimination (all corner cases are caught, no normal samples are flagged), while 0.5 indicates random guessing. Crucially, AUC is

threshold-invariant, meaning it evaluates performance across all possible decision boundaries—vital for real-world deployment, where threshold tuning directly affects false-alarm rates and safety.

However, AUC's strengths also limit its usefulness for corner-case detection, where positive samples (corner cases) are extremely rare. Unlike AUC, which aggregates performance across all thresholds, Average Precision (AP) focuses on the Precision–Recall (PR) curve, which emphasizes the positive class. AP summarizes this curve as a weighted mean of precision values across thresholds, with weights proportional to the increase in recall at each step. This makes AP particularly well-suited for corner-case detection: it directly measures how well a model balances capturing rare risks (high recall) with avoiding false alarms (high precision)—a critical trade-off for deploying safe, reliable systems.

B. Model-Level Metrics

Neuron coverage was first proposed in a white-box differential test algorithm to identify inputs that cause inconsistent outputs across multiple DNNs by applying an affine transformation to the image [47]. Mathematically, $N = n_1, n_2, \dots$ indicates all neurons of a DNN and $T = \mathbf{x}_1, \mathbf{x}_2, \dots$ represents

the test input. The $out(n, \mathbf{x})$ function returns the output of the neuron n . Thus, the coverage of neurons defined is as follows:

$$NCov(T, \mathbf{x}) = \frac{|n| \forall x \in T, out(n, \mathbf{x}) > t|}{|N|} \quad (6)$$

Neuron coverage quantifies the testing of a DNN's internal logic. DeepGauge proposed multi-granularity testing criteria [159], which provide test coverage standards for deep learning systems at two levels: the neuron and layer levels. Accordingly, neurons are divided into two states: the major function region and the corner-case region. For a neuron n , low and $high$ are the upper and lower boundary output values of the activation function range, and if the neuron output is within a reasonable range, this is identified as the major function region. It can be defined as follows:

$$\forall n \in N : \phi(\mathbf{x}, n) \in [low_n, high_n] \quad (7)$$

If it exceeds this range, it is considered a corner-case region.

$$\exists n \in N : \phi(\mathbf{x}, n) \in (-\infty, low_n) \cup (high_n, +\infty) \quad (8)$$

Combinatorial testing (CT) is an effective technique for inspecting traditional software systems, as it balances the trade-off between test coverage and fault detection. DeepCT [160] extends this idea to deep learning systems and proposes a combinatorial testing criterion for DL systems. It introduces the concept of tomography, which treats the DNN as a multilayer (multiscale) function and tests it layer by layer to explore its potential feature space.

C. Semantic-Level Metrics

Semantic-level corner-case detection is predominantly investigated along four principal dimensions: predictive uncertainty, reconstruction error, semantic distance, and distributional shift between datasets. Accordingly, the effectiveness of a given model can be quantitatively assessed using the following metrics.

Structural similarity (SSIM) quantifies the degree of distortion in an image or the resemblance between two images. SSIM is a perceptual model that quantifies the perceptual discrepancy between the original and predicted images. It relies on the image's visible structure and is more consistent with human intuition. The mathematical definition of SSIM is shown in Eq. 9, where μ_x denotes the average gray-scale for measuring brightness, and σ_{xy} denotes the standard deviation of gray-scale for measuring contrast.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (9)$$

In general, mIoU is used to evaluate the performance of semantic segmentation models. Similarly, to detect corner cases at the semantic level, mIoU defined in Eq. 10 is used to quantify the segmentation mask's accuracy. Here TP_t , FP_t , and FN_t are the numbers of true positive, false positive, and false negative pixels in frame t , respectively.

$$IoU_t = \frac{TP_t}{TP_t + FP_t + FN_t} \quad (10)$$

Wilderness Impact (WI) was proposed in [66] as an evaluation index of the corner case. Because unknown objects are prone to misclassification into known categories, WI quantifies the impact of unknown categories on the model's accuracy. Ideally, the smaller the WI value, the less the unknown class interferes with the accuracy.

$$WI = \frac{P_K}{P_{K \cup U}} - 1 \quad (11)$$

where P_K represents the accuracy of the known class in the validation set, and $P_{K \cup U}$ represents the accuracy of the model on the validation set of the known class and the unknown class.

VI. DATASETS

In this section, we summarize publicly available datasets for robust safety-critical corner-case detection.

A. Real-World Datasets

To evaluate the model's performance across various scenarios, extensive real-world data collection has been conducted. As one of the most popular datasets in autonomous driving, KITTI (Karlsruhe Institute of Technology and Toyota Technological Institute) consists of sensory outputs recorded with a variety of sensor modalities, including high-resolution RGB, grayscale stereo cameras, and a 3D laser scanner, and provides 15,000 frames of data around locations in Karlsruhe, Germany. The KITTI dataset is widely used to assess depth prediction, depth completion, and object recognition. Recently, numerous large-scale datasets including nuScenes [161],¹ Cityscapes [177], Waymo [163],² Audi Autonomous Driving Dataset (A2D2) [164]³ and the ONCE (One million sCenEs) dataset for 3D object detection [178] have been collected under various scenarios.

Apart from these general-purpose datasets, a few datasets (such as RoadAnomaly [72], RoadAnomaly21 [179], Lost and Found Dataset [180], Fishyscapes [181]) focus on abnormal or dangerous scenes. For example, RoadAnomaly [72] and RoadAnomaly21 [179] are datasets for anomaly segmentation and can be used to identify regions containing unusual objects (such as animals, rocks, and obstacles) that were never seen during training. DADA [182] can be used to localize video accidents. Although these benchmarks can detect anomalies or predict risky attributes in safety-critical settings, they cannot improve the model's robustness in open, dynamic environments due to poor scenario coverage and limited sample size. To jointly exploit existing datasets, CODA [28] proposed a unified two-stage scheme to select corner cases from large-scale datasets. CODA is a large-scale real-world road corner case dataset for object detection in autonomous driving, consisting of 10,000 carefully selected road driving scenes with image domain tags, well-aligned LiDAR data, and high-quality bounding box annotations for 43 representative object categories. To assess the driving risks posed by critical objects,

¹<https://www.nuscenes.org/>

²<https://waymo.com/open/>

³<https://www.a2d2.audi/a2d2/en.html>

DRAMA [168] is introduced, comprising 17,785 scenario clips recorded from a moving vehicle in highly interactive urban traffic environments.⁴ Meanwhile, it offers a broad range of annotations—including risk-object bounding boxes, free-form captions, and distinct labels for ego-car intention, scene classification, and driver suggestions—which together facilitate understanding of the causes behind risky scenarios.

B. Synthetic Datasets

Due to the limited coverage of real-world data and the substantial costs of data collection [80], synthetic datasets are widely used to capture broader coverage of environmental conditions, traffic scenarios, and driving contexts, while requiring comparatively minimal manual annotation. In contrast to real-world datasets, synthetic datasets can provide large volumes of training data, thereby mitigating dependence on expensive manual labeling of real images and helping address privacy concerns inherent to real data [175]. Nonetheless, they usually target one specific task—for instance, SYNTHIA [170], GTA-V [173], and AWSS [183] are designed for semantic segmentation, while SynPeDS [184] focuses on object detection. Moreover, most of these works rely primarily on domain adaptation techniques to mitigate the domain shift between synthetic and real-world datasets.

Therefore, multitask-generated datasets are gaining popularity. For example, the SHIFT dataset [174] comprises more than 4800 tracking sequences acquired from eight multimodal sensors distributed across different positions, capturing dynamic driving scenes. It enables 13 perception tasks for multi-task-oriented systems, such as semantic/instance segmentation, monocular/stereo depth regression, 2D/3D object detection, 2D/3D multi-object tracking (MOT), optical flow estimation, point cloud registration, trajectory prediction, and human pose estimation. To enrich scene diversity, MUAD [175], a generated dataset for autonomous driving with multiple types of uncertainty and tasks, is proposed. Specifically, The MUDA dataset contains 3,420 training images, 492 validation samples, and 6,501 test samples. Each split includes both day and night conditions, with two-thirds of the images captured during the day and one-third at night. The test sets additionally span diverse weather conditions (rain, snow, and fog at two intensity levels) and include multiple out-of-distribution objects [175].⁵

Although numerous generated datasets are available, the data distributions of generated images are significantly different from those of real-world images. Domain drifts make it impractical to apply the model trained on generated datasets directly to real-world scenarios [185], [186]. Recently, an open-source photorealistic simulator for autonomous driving was released to enable the construction of realistic environments for the training and testing of autonomous vehicles [151]. Apollo Synthetic⁶ is a photorealistic synthetic dataset for autonomous driving, comprising 273k unique samples captured across a wide range of high-visual-fidelity virtual scenes,

including highways, urban areas, residential neighborhoods, downtown districts, and indoor parking garages.

VII. CHALLENGES AND OPPORTUNITIES

A. Building Large-scale Datasets of Corner Cases

Although many autonomous-driving datasets exist, few explicitly target corner cases. Given the strict safety requirements of autonomous systems, there is an urgent need for large-scale datasets focused on rare, safety-critical scenarios. First, heterogeneous and fragmented data formats hinder reuse. Existing datasets use incompatible schemes and annotation protocols, preventing systematic integration of corner-case samples. A unified data schema is needed to integrate dispersed public datasets and reduce redundant data collection. Second, severe class imbalance undermines model robustness. Existing datasets disproportionately emphasize nominal driving conditions while insufficiently capturing rare and extreme scenarios, thereby diminishing system fault tolerance and perpetuating a discrepancy between highly specialized corner-case handling and robust responses to abrupt, high-risk events. Large-scale corner-case datasets are therefore needed to cover diverse challenging contexts. The SDAC dataset [187], designed for anomaly detection with dense multimodal annotations, effectively complements real-world operational data. Third, traditional centralized collection pipelines struggle to capture rare extreme events. Crowdsourcing strategies [188], deployed via civilian vehicle fleets and platforms such as CrowdOS [189], can assign multimodal data-collection tasks along regular routes, offering a scalable, cost-effective way to acquire corner-case data.

B. Multi-Modality Detection of Corner Cases

In safety-critical corner cases of autonomous driving, the capabilities of current perception systems remain fundamentally constrained. Most existing methods operate on single-modal input and thus fail to fully exploit the complementary information provided by multi-sensor configurations (e.g., cameras, radar, and LiDAR). Each modality exhibits distinct advantages and limitations in such scenarios: RGB imagery supplies semantically rich and fine-grained visual information but is highly sensitive to illumination changes and adverse weather conditions [7], [190], [191], [192]; LiDAR delivers accurate geometric and depth measurements yet suffers from spatial sparsity and performance degradation under adverse environmental conditions [193], [194]; radar is inherently robust to interference and challenging weather, but its spatial resolution is comparatively low. To mitigate these limitations, the research community has increasingly focused on multi-sensor fusion strategies. Representative paradigms include point-level fusion [195], [196], feature-level interaction across modalities [197], [198], [199], [200], [201], [202], and fusion via projection into a unified representation space, with Bird's-Eye-View (BEV) becoming particularly prominent [203], [204], [205], [206]. For instance, BEVFusion projects both camera images and LiDAR point clouds into a shared BEV space to enable joint reasoning [204], while BEVGuide proposes a universal BEV framework designed to generalize

⁴<https://usa.honda-ri.com/drama>

⁵<https://muad-dataset.github.io/>

⁶<https://developer.apollo.auto/synthetic.html>

across different sensor setups and tasks [203]. In parallel, camera-only multi-view 3D object detection has made substantial progress through the incorporation of auxiliary prototypes and contrastive objectives, which improve robustness under occlusion and inclement weather conditions [207].

Despite these advances in multi-modal and single-modal perception for corner cases, current research faces three primary bottlenecks: 1) Dedicated datasets annotated with “modality failure–scenario mutation” labels are scarce, and the effective integration of real-world and synthetic data remains challenging, which restricts the training and validation of multi-modal corner-case detection models. 2) Datasets supporting multimodal corner-case analysis are extremely scarce, leading to the poor performance of existing multimodal fusion methods in extreme corner-case scenarios, especially in single-modality failure detection. 3) Insufficient cross-modal semantic alignment commonly exists, which causes significant segmentation performance degradation in low-contrast regions of corner-case scenes. Recent studies have introduced a range of promising approaches to address these challenges. T-READi [208] enhances robustness under modality failure, while roboFormer [209] improves cross-modal alignment. The corner prediction model further reinforces the complementarity between perception and prediction [210]. For example, JiSAM leverages synthetic data in conjunction with scarce labeled real-world data to mitigate data scarcity and reduce the sim-to-real gap in LiDAR perception [211]. Moreover, active modality selection [212], [213], [214] holds additional promise for autonomous driving perception by adaptively configuring sensor combinations according to the driving context, thereby improving accuracy while simultaneously reducing computational overhead [215], [216].

C. Corner Case Detection in Mixed Driving System

Nowadays, vehicles with partial automation have already been widely available on the market, and the full automation era with “driverless” capabilities is near the horizon. In the mixed driving system, corner case detection faces challenges at both the individual and the system level.

Partially automated vehicles, conceptualized as human–machine systems, must perceive environmental hazards, quantify risks, and initiate interventions in rare, safety-critical situations [217]. Reliable assessment of sensor outputs is essential; if measurement or interpretation cannot be trusted, the control authority should be transferred to the human driver. This requirement adds complexity by introducing uncertainty from stochastic human behavior and diverse human–automation interaction modes. Current active shared-control paradigms struggle with robust driver-intent inference, accurate estimation of situational awareness, and formal modeling of driver trust, hindering safe and timely control handovers [217]. To mitigate these issues, an end-to-end learning-based architecture within a modular control framework has been proposed to automatically flag corner cases for faster, more focused reliability assessments [218]. Complementarily, a human-vision-inspired perception pipeline on the “Tianmou” neuromorphic vision chip enables low-latency sensing and processing, thereby enhancing the

responsiveness and robustness of control transfers [219]. In addition, new methods extract rare, safety-critical scenarios from large-scale urban intersection data and convert them into simulation-ready test cases, advancing validation and verification of automated driving under real-world conditions [220].

In hybrid traffic systems [221], distributed sensor networks face challenges in fully exploiting vehicle heterogeneity. AutoFed applies federated learning to aggregate data across vehicles [222]. To enhance safety, the MPC framework of Mohseni et al. [223] and the game-theoretic system of Hang et al. [224] are proposed to support cooperative maneuvers. Hybrid TrafficAI [225] enhances both anomaly detection and decision-making in hybrid traffic environments. Moreover, a lightweight deep learning model tailored for corner radar provides robust real-time speed estimation, resolving Doppler ambiguity and mount-angle constraints without requiring additional sensors [226]. Overall, progress in corner-case detection requires, at the individual level, robust and trustworthy human–machine interaction, and, at the system level, distributed cooperative frameworks that effectively integrate theoretical advances with practical deployment.

D. Photolistic Corner Case Generation

Numerous image synthesis solutions have been studied, but few can be applied to synthesize corner cases in autonomous driving. First, models trained predominantly on synthetic data frequently underperform in real-world settings because of domain discrepancies in illumination, texture, and scene dynamics [227]. Sim-to-real adaptation techniques can partially mitigate these issues by aligning known and unknown domains [228]. In this context, DriveArena [229] offers a promising approach by generating data closely aligned with real-world distributions, enabling more seamless deployment of trained agents. Second, generic image generation approaches—such as text–sketch fusion [230] and bounding box–text grounding [231]—do not offer sufficient semantic or behavioral granularity for complex, dynamic traffic environments. HumanSim [232] partially addresses this limitation by simulating human-like driving, explicitly modeling spatial relationships, fine-grained agent interactions, and transparency–efficiency trade-offs, thus enabling more realistic dynamic corner cases. Third, many generated scenarios still lack the stochastic behavioral patterns and heterogeneity characteristic of real-world traffic. Conventional methods [227], [228], [230], [231], [233], [234] typically focus on controlling object placements rather than modeling underlying intent, decision-making processes, or socio-cultural context. HumanSim [232] only partially captures human decision-making heuristics, whereas DriveArena [229] primarily focuses on generating diverse global traffic flows. Both approaches, therefore, require integration with region-specific behavioral data and cultural priors to more comprehensively represent driver intent and heterogeneous driving norms. Furthermore, in terms of complex safety-critical corner cases, current research lacks a universal framework for generating collision trade-off scenarios, and pays limited attention to the ethical constraints in such scenario generation. This ethical and scenario diversity gap

remains a key issue to be addressed in photorealistic corner case generation research.

E. Large Language Models for CC Understanding

With the remarkable advancements of large language models (LLMs) across diverse domains, their integration into autonomous driving systems has garnered significant traction—enriching human-machine interaction [235], [236], [237], enhancing environmental perception capabilities [238], [239], and enabling a paradigm shift from data-driven to knowledge-driven end-to-end pipelines [240], [241], [242], [243]. For instance, Cui et al. [244] and Yang et al. [245] have presented comprehensive literature reviews on the application of LLMs in autonomous driving, synthesizing existing progress in this rapidly evolving field. However, leveraging LLMs to effectively understand and address corner cases—a critical bottleneck for safe and reliable autonomous navigation—remains in its nascent stages, plagued by several unresolved challenges.

First, while a plethora of visual language models (VLMs, e.g., VisionLLM [246]) and multimodal LLMs (MLLMs, e.g., LLaVa [247]) have been developed, these models are inherently general-purpose and lack specialized optimization for autonomous driving scenarios, particularly extreme cornering contexts. To address this, Chen et al. proposed CODA-LM [248], leveraging a hierarchical data structure and LLM prompting to generate high-quality pre-annotations for complex scenes. CODA-REC [249] augmented with textual descriptions to facilitate MLLM-based comprehension of multimodal semantic information—addressing the inherent limitation of general-purpose models in interpreting domain-specific corner scenarios. Second, while existing studies have sought to integrate VLMs for automated decision-making in autonomous driving, they rarely prioritize robustness and interpretability. Mo et al. [250] fine-tuned InternVL-2.0 for driving scenarios by enhancing spatial correlation with position and depth information, incorporating chain-of-thought reasoning to improve accuracy, and developing context-aware learning via scenario-based retrieval. However, it still highlights the need for further refinement to fully address corner-case complexity. Furthermore, integrating LLMs and VLMs into autonomous driving systems remains challenging due to hallucination risk and real-time performance limitations.

VIII. CONCLUSION

Understanding corner cases is critical for decision-making in safety-critical scenarios. However, research on corner cases in autonomous driving has received comparatively less attention, overshadowed by topics like OOD learning, anomaly detection, and adversarial examples. In this survey, we conduct a taxonomic analysis of data samples and their relationships to decision boundaries. We observe that corner cases remain understudied due to three key gaps: limited consensus in their definition, a scarcity of dedicated datasets, and diverged research priorities.

To accelerate the development of autonomous driving, both academia and industry should make greater efforts across

multiple dimensions to deepen understanding of corner cases. First, large-scale corner-case datasets, either collected in the real world via crowdsourcing or generated by deep generative neural networks, are prerequisites. Second, how to fuse multimodal outputs for corner-case detection in the mixed driving system is promising for improving model performance, especially in complex scenarios. Third, the current data-driven paradigm has substantial performance limitations, including low robustness, limited interpretability, and poor fault tolerance. With the popularity of LLMs, leveraging prior knowledge to improve corner-case understanding and building a knowledge-based decision-making system are promising approaches for the safety of autonomous driving.

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